Exponentials & Logarithms

Section 1: The Natural Exponential & Logarithm

1 Solve the equation $e^{2x-1} = 5$, giving your answer in exact form.

a. $x = \dfrac{1 + \ln 5}{2}$

b. $x = \dfrac{\ln 5 - 1}{2}$

c. $x = \ln\left(\dfrac{5}{2}\right) + 1$

d. $x = \dfrac{\ln 6}{2}$

2 Solve $\ln(3x + 2) - \ln(x) = \ln 4$ for $x > 0$.

a. $x = 2$

b. $x = 1$

c. $x = 4$

d. $x = 1/2$

3 Simplify the expression $e^{3\ln x} + \ln(e^{2x})$ for $x > 0$.

a. $x^3 + 2x$

b. $3x + 2x = 5x$

c. $x^3 + e^{2x}$

d. $3x^2 + 2$

Section 2: Exponential Equations with Different Bases

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4 Solve the equation $2^{x+1} = 3^x$, giving x in terms of natural logarithms.

a. $x = \dfrac{\ln 2}{\ln 3 - \ln 2}$

b. $x = \dfrac{\ln 3}{\ln 3 - \ln 2}$

c. $x = \dfrac{\ln 2}{\ln 3 + \ln 2}$

d. $x = \dfrac{1}{\ln(3/2)}$

5 Solve the equation $e^x - 6e^{-x} = 1$ for real x.

a. $x = \ln 3$

b. $x = \ln 3$ or $x = \ln(-2)$

c. $x = \ln 2$

d. $x = 3$

6 Solve $5^{2x-1} = 7$, giving your answer correct to 3 significant figures.

a. 1.10

b. 0.852

c. 1.21

d. 0.945

Section 3: Transforming to Linear Form ($y = Ax^n$ and $y = Ab^x$)

7 The variables x and y satisfy $y = ab^x$. When $\ln y$ is plotted against x, a straight line is obtained. What are the gradient and vertical intercept?

a. Gradient $= \ln b$, Intercept $= \ln a$

b. Gradient $= b$, Intercept $= a$

c. Gradient $= \ln a$, Intercept $= \ln b$

d. Gradient $= a$, Intercept $= \ln b$

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8 The variables x and y satisfy $y = kx^n$. The line of $\ln y$ against $\ln x$ has gradient 2.5 and intercept $\ln y = 1.386$. Find k and n.

a. $k = 4, n = 2.5$

b. $k = 2.5, n = 4$

c. $k = 1.386, n = 2.5$

d. $k = e^{2.5}, n = 1.386$

9 Points $(2, 3.2)$ and $(5, 5.6)$ lie on the straight line of $\ln y$ against x. Find the value of b if $y = ab^x$.

a. $b = e^{0.8} \approx 2.226$

b. $b = 0.8$

c. $b = 2.4$

d. $b = e^{1.6} \approx 4.95$

Section 4: Exponential Growth and Decay Modeling

10 The mass of a radioactive substance decays according to $M(t) = 80e^{-0.05t}$, where t is in years. Find the half-life of the substance in years.

a. ≈ 13.9 years

b. 20.0 years

c. 10.0 years

d. 15.5 years

11 A bacterial culture grows according to $N(t) = 500e^{kt}$. If $N(3) = 2000$, find the exact value of k.

a. $k = \dfrac{\ln 4}{3}$

b. $k = \dfrac{\ln 2}{3}$

c. $k = \dfrac{4}{3}$

d. $k = \ln\left(\dfrac{4}{3}\right)$

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12 A hot beverage cools according to Newton's law of cooling: $T(t) = 20 + 70e^{-0.08t}$, where T is temperature in $^\circ$C and t is time in minutes. How long does it take to cool to $45\,^\circ$C?

a. ≈ 12.9 min

b. 15.2 min

c. 10.5 min

d. 18.0 min

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